

Wenshu Zhao

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EDUCATION

University of Washington, Seattle	Sep.2024 - Jun.2026(Expected)
M.S. Information Science (Specialization: Data Science & Artificial Intelligence)	GPA: 3.9/4.0
China Pharmaceutical University, Nanjing	Sep.2020 - Jun.2024
B. S. Information Management and Information Systems	GPA: 3.7/4.0

SKILLS

Programming Languages: Java, Python, JavaScript, SQL, C++, TypeScript, HTML/CSS

Frameworks & Tools: Spring Boot, React, FastAPI, Hadoop, Apache Spark, Git, Expo, Xcode

Databases & Cloud: PostgreSQL, MySQL, MongoDB, SQL Server, Redis, AWS, Azure, Docker, Linux, CI/CD, Nginx

Concepts: Distributed Systems, Microservices, System Design, Data Structures, SDLC

AI Tools: Claude, GitHub Copilot, Cursor, ChatGPT (Prompt Engineering, Agent Workflow Design, Code Generation)

PROFESSIONAL EXPERIENCE

Software Engineer - Capstone - Voraa Sep.2025 - Present

- Designed and implemented Deal-Breaker Mode across **Flutter** and **Firebase Cloud Functions** using **Node.js**, executing a multi-constraint filtering pipeline against a dynamic GenTag taxonomy for real-time venue matching.
- Solved Firestore's single array-contains limitation by implementing in-memory multi-tag AND/OR filtering with pagination and fallback relax suggestions on empty results.

Software Engineer Intern - IpserLab Jun.2025 - Dec.2025

- Architected a schema-driven form engine in **React Native & TypeScript** supporting 6-step workflows with 28+ dynamic fields, decoupling UI from business logic via reusable components and custom hooks.
- Engineered an offline-first application lifecycle system managing draft-to-completion submission state via **AsyncStorage** with per-user data namespacing, ensuring cross-account data isolation.
- Designed a Command pattern API routing system in **Jakarta EE**, mapping endpoint handlers to URL paths as strategy interfaces to enable extensible routing without modifying core logic.
- Implemented RESTful APIs in **Java 21 / JDBC / PostgreSQL** for multipart form submissions and binary file storage, integrating a React Native mobile client with backend services across a full-stack application.

Software Engineer Intern - Legend Biotech Feb.2023 - Apr.2023

- Developed and launched an internal attendance system (**Java/JavaScript**) integrated with WeCom, used by 500+ employees, improving attendance tracking efficiency by 40%.
- Engineered patient data management modules for a CAR-T treatment tracking system using **Spring Boot** and improved frontend user experience with **React**, enabling real-time treatment schedule updates and progress tracking for 200+ patients, in collaboration with clinical staff and full-stack engineers.

PROJECT EXPERIENCE

Seattle Source Ranker Sep.2025 - Dec.2025

- Built a distributed ingestion system with **Celery & Redis** to collect and validate **400K+** repositories from **28K** Seattle developers, achieving **8.1×** throughput improvement and **99.98%** ingestion reliability under rate limits.
- Architected a fault-tolerant API orchestration layer combining **GraphQL** and **REST fallback** strategies to correct inconsistent watcher metrics, eliminating ~2% invalid repositories and ensuring ranking data integrity.
- Implemented a rate-limit-aware token rotation system with adaptive backoff and idempotent retries, reducing rate-limit failures by 92% and enabling full dataset rebuilds within 60-90 minutes.
- Automated daily dataset updates and static deployment via GitHub Actions (**Bash** scripts), generating 10K+ optimized JSON datasets consumed by a React interface with real-time search and sub-50ms page load times.

Virtual Character Letter Platform Nov.2024 - Apr.2025

- Built a Python **FastAPI** service for LLM-powered letter generation with dynamic character-specific prompts, multi-language support and multi-turn context, integrating **GPT** and **Doubao** via OpenAI-compatible APIs.
- Refactored LLM integration using the **Strategy Pattern**, enabling zero-change backend extension; added exponential backoff retry with 4xx short-circuit to eliminate redundant retries on auth failures.
- Architected a two-layer async system to decouple Java-Python interactions: **FastAPI BackgroundTasks** for non-blocking HTTP responses, and **RabbitMQ** for persistent queued task delivery with automatic retry on failure.
- Reduced latency and supported flexible data modeling by implementing **Redis** caching and a multi-database architecture, using **MongoDB** for unstructured letter content and **MySQL** for structured user and role data.
- Ensured reliable deployment and operational stability by containerizing services with **Docker**, deploying to **Amazon EC2**, configuring **Nginx** as a reverse proxy, and setting up **AWS CloudWatch** for monitoring.